How Faithful Is the Circuit You Found? Post-Selection-Valid Confidence Bounds for Mechanistic Interpretability

Anonymous

Abstract

Circuit papers end with a number: "our circuit recovers 87% of the model's performance." That number is almost always the largest of many, because the circuit was chosen by maximising it on the very interventions used to score it. A maximum over a search is biased upwards, and an error bar computed as though no search had happened is too narrow: for 100 equally good hypotheses the nominal 95% interval covers 0.6% of the time.

We treat graded faithfulness — interchange-intervention accuracy, logit difference recovered, behavioural agreement under ablation — as a population quantity, treat circuit search as a selection rule, and supply lower confidence bounds that stay valid however the circuit was chosen. We prove that the inflation of a selected score is of order $\sigma\sqrt{\log|\mathcal{C}|/n}$ in the size of the searched class and the number of interventions, and cannot be argued away. We then give five corrections — an empirical-Bernstein union bound, a prior-weighted (Occam) bound that handles all 2^{144} subsets of GPT-2 small's attention heads and makes small circuits cheaper to certify than large ones, an empirical bootstrap-t simultaneous band that adapts to the strong correlation between overlapping circuits, sample splitting, and selective intervals for the winner — and validate their coverage exactly on four tiny transformers with known algorithms.

On GPT-2 small's indirect-object-identification circuit, searching 840 circuit hypotheses with $n = 1000$ interventions, the naive interval covers 59% of the time; the bootstrap-t band covers 99% and still certifies faithfulness of 0.992, against a (inflated) naive point estimate of 0.998. The correction is therefore cheap — but only if two other things are done right, and both are usually not. First, if the target population is new prompt *templates* rather than new fills of the same ones, the effective sample size shrinks by a factor of 4.6 and the naive interval's coverage falls to 73% *and keeps falling as more prompts are added.* Second, if the search was adaptive, a bound uniform over the circuits the search could have reached is valid but vacuous, and interventions must be held out instead. We release a library that turns any per-instance faithfulness score into a certificate.

1 Introduction

A mechanistic interpretability paper typically ends with a sentence of the form "our circuit recovers 87% of the model's performance". Behind that sentence there is a search. The investigator considered many candidate circuits — which attention heads to keep, which subspace of a layer's activations to align with which variable of the hypothesised algorithm, which token positions to intervene at — scored each of them on a set of interventions, and reported the best. The number that reaches the reader is therefore a maximum over a search, not a measurement of a hypothesis fixed in advance. Maxima of noisy quantities are biased upwards, and standard errors computed as though no search had happened are too small. **This paper supplies the missing warranty: lower confidence bounds on faithfulness that remain valid no matter how the circuit was chosen, together**

with a measurement of what the usual practice costs. The problem is the winner's curse, and it is the same one that motivated valid post-selection inference in statistics [5, 33], the reusable holdout in adaptive data analysis [18], and the study of leaderboard overfitting in machine learning [7, 41, 42].

Mechanistic interpretability has good reason to care. *Causal abstraction* [26] — the framework that turns "this algorithm describes this network" into a testable statement, by requiring the two to agree under matched interventions — gives the field a precise notion of what a circuit hypothesis claims, and a graded version of it (how *often* the network's behaviour under an intervention matches the algorithm's) is what circuit papers actually report. If that number is systematically optimistic, then so is the field's picture of how well it understands its models. And the searches involved are large: automated circuit discovery sweeps thousands of subgraphs [16, 46], distributed alignment search optimises over rotations of the residual stream [25], and a manual analysis of a single behaviour can compare dozens of hypotheses [49].

What this paper does. We treat graded faithfulness as a population quantity, treat circuit search as a selection rule, and ask for confidence bounds that survive the selection.

1. **We quantify the optimism** (Section 3). For a class of m hypotheses evaluated on n interventions, the expected inflation of the reported score is at most $\sigma\sqrt{2\log m/n}$ and, in the worst case, at least a constant times $\sigma\sqrt{\log m/n}$ when the hypotheses are independent, so the worst case cannot be improved by better analysis. We also give a closed form for the coverage of the naive interval, which depends sharply on how correlated the hypotheses are.

2. **We give five post-selection-valid constructions** (Section 4): a distribution-free empirical-Bernstein union bound; a prior-weighted ("Occam") version that handles classes as large as all subsets of the 144 attention heads of GPT-2 small, and under which small circuits are cheaper to certify than large ones; a bootstrap-t simultaneous band that adapts to the correlation between overlapping circuits; sample splitting, which is the right answer when the search is adaptive; and selective (conditional) intervals for the winner, which we find help only when the winner wins clearly.

3. **We identify the unit of replication as a first-order issue** (Section 5). Interpretability datasets are generated from a small number of prompt templates. Resampling prompts answers a question about new fills of the same templates; resampling templates answers a question about the behaviour itself. The two differ by a *design effect* — the factor by which clustering inflates the variance of an average — which we measure to be 4.6 on the indirect-object-identification (IOI) task.

4. **We validate coverage exactly** (Section 6) on four tiny transformers trained to 100% accuracy on algorithmic tasks — one of which has a known-faithful alignment planted by interchange intervention training — and then on GPT-2 small's IOI circuit. Because we cache a per-instance score matrix once and treat the cached pool as the population, the true faithfulness of every hypothesis is known exactly and measured coverage carries no oracle noise.

5. **We give a protocol** (Section 7) and a library that converts any per-instance faithfulness score into a certificate.

Why this is not just statistics with a new application. Four things about faithfulness evaluation are not generic, and each forced a change to the standard machinery. *(i)* The dominant scores are binary (did the intervened network agree with the algorithm?), and for binary scores the sample mean and sample standard deviation are dependent, which breaks the textbook multiplier bootstrap; we had to recompute the variance inside every replicate to recover nominal coverage (Section 4(c)). *(ii)* The most widely reported metric is a ratio of means, not a mean, so the finite-sample bounds do not apply to it at all and it needs its own influence-function band (Section 4.1). *(iii)* Circuit classes are power sets of model components, which is what makes a size-stratified prior the natural device and yields the interpretability-relevant conclusion that *minimal circuits are cheaper to certify.* *(iv)* The unit of replication is a prompt template, not a prompt, and choosing between the two changes the effective sample size by more than any of the corrections do (Section 5). The empirical quantities we report — how much a circuit search inflates its own score, how many hypotheses a class is *effectively* worth, how correlated circuits are — are measurements about interpretability practice, not about statistics.

What this paper is not. We do not propose a new faithfulness metric, a new circuit-discovery algorithm, or a new notion of what a circuit is. We take the field's metrics and searches as given and ask what can honestly be claimed from them. Nor do we claim that published circuits are wrong: a bound that is valid after selection is usually only modestly below the naive estimate (Section 6), and our GPT-2 results certify a genuinely faithful IOI circuit. The claim is that the guarantee has to be earned, and that earning it is cheap.

2 Setup: graded faithfulness as a population quantity

Interventions as data. Let Z denote everything random about a single faithfulness measurement. In an interchange-intervention experiment [23], $Z = (b, s)$ is a pair of inputs: a *base* b and a *source* s. In an ablation experiment, Z is a single input. We assume $Z_1, \ldots, Z_n \sim P$ independently; Section 5 relaxes this to clustered sampling and explains why the relaxation matters.

Hypotheses and scores. A *circuit hypothesis* c is any object that, together with Z, determines a bounded per-instance score $\varphi(c; Z) \in [0, 1]$. Two instances cover most of the literature:

(i) Alignment hypotheses. c names a location in the network (a component, a token position) and a linear subspace of the representation there, and claims that this subspace carries a high-level variable V. The score is the interchange-intervention indicator

$$\varphi(c; (b, s)) \;=\; \mathbf{1}\Big\{\, f_{\text{low}}\big(b\,;\,\text{subspace} \leftarrow \text{its value on } s\big) \;=\; f_{\text{high}}\big(b\,;\,V \leftarrow V(s)\big)\Big\},$$

whose mean is *interchange-intervention accuracy* [24, 25, 51].

(ii) Component-subset hypotheses. c is a set of model components; every component outside c is ablated (here: replaced by its mean over a reference distribution), and φ records whether the ablated model still behaves like the full model [11, 16, 49].

We write $\theta(c) = \mathbb{E}_P[\varphi(c; Z)]$ for **graded faithfulness** and $\hat{\theta}_n(c) = n^{-1} \sum_i \varphi(c; Z_i)$ for its estimate. Section 4.1 extends everything to the normalised logit difference recovered, which is a ratio of means rather than a mean.

Search. A *selection rule* is any measurable map $\hat{c} = A(Z_1, \ldots, Z_n)$ into the hypothesis class $\mathcal{C}$. Nothing below requires A to be the arg-max: it may be a greedy pruner, a gradient-based alignment search, or an investigator looking at a plot.

Definition 1 (Post-selection validity)**.** A data-dependent function $L(\cdot)$ is a *post-selection-valid* lower confidence bound at level α if $\mathbb{P}\big(\theta(\hat{c}) \geq L(\hat{c})\big) \geq 1-\alpha$ for *every* selection rule $\hat{c}$ that is measurable with respect to the data used to build L.

The cleanest sufficient condition, and the one all our band-type methods satisfy, is simultaneity: $\mathbb{P}(\forall c \in \mathcal{C} :\ \theta(c) \geq L(c)) \geq 1-\alpha$. Simultaneity is strictly stronger than what is needed, but it buys something useful: it certifies every circuit the investigator looked at, including an entire reported Pareto frontier, not just the winner.

3 The problem: how optimistic is a selected faithfulness score?

Proposition 1 (Upper bound on the optimism)**.** *Let $|\mathcal{C}| = m$, $\sigma^2_{\max} = \max_c \operatorname{Var} \varphi(c; Z)$ and let $\hat{c}$ be any selection rule. Then*

$$\mathbb{E}\big[\hat{\theta}_n(\hat{c}) - \theta(\hat{c})\big] \ \leq\ \mathbb{E} \sup_{c\in\mathcal{C}} \big(\hat{\theta}_n(c) - \theta(c)\big) \ \leq\ \sigma_{\max}\sqrt{\frac{2\log m}{n}} + \frac{\log m}{3n}.$$

Proposition 2 (The rate is attained)**.** *Suppose the centred scores are i.i.d. $N(0, \sigma^2)$ across instances and independent across hypotheses, and $\theta(c) \equiv \theta_0$. Then for $\hat{c} = \arg\max_c \hat{\theta}_n(c)$,*

$$\mathbb{E}\big[\hat{\theta}_n(\hat{c}) - \theta(\hat{c})\big] \ =\ \frac{\sigma}{\sqrt{n}}\,\mathbb{E}\Big[\max_{j\leq m} N_j\Big] \ \geq\ \frac{\sigma}{\sqrt{2}}\sqrt{\frac{\log m}{n}}, \qquad m \geq 3.$$

Proofs are in Appendix A. Together the two propositions pin the optimism at order $\sigma\sqrt{\log m/n}$: it cannot be argued away by a better analysis, only reduced by collecting more interventions or searching a smaller class. The dependence on the class size is mild — a hundredfold larger search inflates the optimism by only about 40% when $m \approx 100$, because $\sqrt{\log m}$ grows slowly — but the dependence on n is the usual $n^{-1/2}$, so the optimism does not vanish at any sample size an interpretability experiment is likely to use. What the propositions do *not* pin down is the constant in a given study, because that depends on σ and on how correlated the hypotheses are. Correlation is the crucial study-specific quantity, as the next result shows.

Proposition 3 (Exact coverage of the naive interval under the complete null)**.** *Suppose $\theta(c) \equiv \theta_0$ and the standardised estimates $G_c = \sqrt{n}(\hat{\theta}_n(c) - \theta_0)/\sigma(c)$ are jointly Gaussian with equicorrelation ρ, with $\hat{\sigma}$ treated as known (exact in the Gaussian limit). The one-sided naive bound $\hat{\theta}_n(\hat{c}) - z_{1-\alpha}\sigma(\hat{c})/\sqrt{n}$ at the arg-max covers if and only if $\max_c G_c \leq z_{1-\alpha}$, so its coverage is*

$$\mathbb{E}_{W\sim N(0,1)}\Big[\Phi\big((z_{1-\alpha} - \sqrt{\rho}\,W)/\sqrt{1-\rho}\big)^m\Big],$$

which equals $\Phi(z_{1-\alpha})^m$ at $\rho = 0$.

Figure 1(a) plots this. At $m = 100$ the nominal 95% bound covers 0.6% of the time when hypotheses are independent and 81.3% when they are 90% correlated. Circuit hypotheses *are* strongly correlated — overlapping circuits share most of their components, so their per-instance scores move together — which is why the situation in practice is bad but not catastrophic, and why a method that adapts to the correlation is worth having.

4 Bounds that survive the search

All five constructions below take the same input: the $n \times m$ matrix $S_{ic} = \varphi(c; Z_i)$ of per-instance scores. Computing it once is the only model-forward cost; everything after is arithmetic.

(a) Empirical-Bernstein union bound (union-eb). For $X_i \in [0,1]$ i.i.d., Maurer and Pontil [39] show that with probability at least $1-\delta$, $\mathbb{E}X \geq \bar{X} - \sqrt{2V_n \log(2/\delta)/n} - 7\log(2/\delta)/(3(n-1))$, where V_n is the sample variance. Taking $\delta = \alpha/m$ and a union bound gives a simultaneous band. It is distribution-free and finite-sample, so it is valid for *any* selection rule, and its variance adaptivity matters here: a faithful circuit scores close to 1 on almost every instance and so has tiny variance. Hoeffding's bound is tighter when scores sit near the middle of the range; running both at $\alpha/2$ and taking the larger bound is valid and removes the need to guess (UNION-BEST).

(b) Prior-weighted (Occam) bound (occam). Replace α/m by $\alpha\pi(c)$ for any prior π on $\mathcal{C}$ fixed before seeing data:

$$\theta(c) \;\geq\; \hat{\theta}_n(c) - \sqrt{\frac{2V_n(c)L_c}{n}} - \frac{7L_c}{3(n-1)}, \qquad L_c = \log\frac{2}{\alpha\pi(c)},$$

simultaneously over $\mathcal{C}$. With the size-stratified prior over subsets of N components, $\pi(c) = \big[(N{+}1)\binom{N}{|c|}\big]^{-1}$, so the price a circuit pays grows with $\log\binom{N}{|c|}$. At $N = 144$ (the attention heads of GPT-2 small) that is 34.3 nats for a 10-head circuit and 97.1 nats for a 72-head circuit; at $n = 2000$ interventions and a score with standard deviation 0.3 the two bounds sit 0.10 and 0.21 below the point estimate. *Small circuits are statistically cheaper to certify*, which aligns the statistics with the field's preference for minimal circuits. This is the only one of the five that applies when the class is the full power set of components.

(c) Bootstrap simultaneous band (boot-max). Resample the n interventions with replacement, recompute *both* the mean $\hat{\theta}^*_n(c)$ and the standard deviation $\hat{\sigma}^*(c)$ of every hypothesis, and let $\hat{q}_{1-\alpha}$ be the $(1-\alpha)$ quantile of the studentised maximum $\max_c \sqrt{n}\,(\hat{\theta}^*_n(c) - \hat{\theta}_n(c))/\hat{\sigma}^*(c)$. Then

$$L(c) \;=\; \hat{\theta}_n(c) - \hat{q}_{1-\alpha}\,\hat{\sigma}(c)/\sqrt{n}$$

is an asymptotically valid simultaneous band even when m grows much faster than n [15, Thm. 4.3]; the non-studentised counterpart is Chernozhukov et al. [12, Cor. 3.1], and Appendix A states the conditions. Unlike Bonferroni it adapts to the dependence between hypotheses.

One failure mode, and its fix. The band is studentised, so a hypothesis that scores perfectly on *every sampled* instance has $\hat{\sigma}(c) = 0$ and would be certified at its point estimate — no resampling of those instances can discover that its population value is below one. We therefore floor the studentising standard deviation at $n^{-1/2}$ (in the observed statistic and in every replicate alike), which gives such a hypothesis an interval of width $\hat{q}_{1-\alpha}/n$, the order of the exact "rule of three". For classes that may contain such hypotheses we also provide BOOT-FLOORED: use the finite-sample bound of (a) for the degenerate columns and the band for the rest, each at level $\alpha/2$. Both ingredients are simultaneous over the whole class, so a union bound over their two failure events makes any data-dependent choice between them valid at level α.

Recomputing the standard deviation is not optional. The textbook multiplier bootstrap holds $\hat{\sigma}(c)$ fixed at its observed value. For the *binary* scores that dominate interpretability, the sample mean and the sample standard deviation are strongly dependent — for a score above 1/2, an upward deviation of the mean necessarily comes with a smaller standard deviation — so the true studentised maximum has a heavier upper tail than a fixed-$\hat{\sigma}$ bootstrap can reproduce. In our simulations with $m = 60$ hypotheses and $n = 400$ interventions the fixed-$\hat{\sigma}$ band covers 89% instead of 95%, while the version above covers 95%. We therefore use the empirical (resampling) bootstrap-t throughout, and the same correction applies to the cluster-robust version of Section 5. We summarise the adaptivity to dependence by the

Definition 2 (Multiplicity factor and effective number of hypotheses)**.** Let $\hat{q}^{\text{marg}}_{1-\alpha}$ be the critical value the *same* bootstrap gives for a single, pre-specified hypothesis. The *multiplicity factor* is $\kappa = \hat{q}_{1-\alpha}/\hat{q}^{\text{marg}}_{1-\alpha}$ — the factor by which searching widens the interval — and the *effective number of hypotheses* is $m_{\text{eff}} = \alpha/\big(1 - \Phi(\kappa\, z_{1-\alpha})\big)$, the number of independent hypotheses a Bonferroni correction would have to cover to pay the same price.

Dividing by the marginal critical value matters: $\hat{q}_{1-\alpha}$ also absorbs the finite-sample non-normality of the studentised statistic, which has nothing to do with searching. The natural comparison for κ is the Bonferroni factor $z_{1-\alpha/m}/z_{1-\alpha}$ that a correction over m *independent* hypotheses would pay. In our simulations (Table 3) the two are close when hypotheses are independent and κ falls well below Bonferroni once they are correlated — at $|\mathcal{C}| = 2000$ and correlation 0.9, $\kappa = 1.99$ against a Bonferroni factor of 2.47. κ can also *exceed* the Bonferroni factor, and does so on IOI, where many circuits are perfect on every sampled instance and the studentised maximum is correspondingly heavy-tailed; that is a finite-sample effect, not multiplicity, and it is the situation the floored variant of paragraph (c) is for.

(d) Sample splitting (split). Select on a fraction f of the instances and bound on the rest. Valid for arbitrary and *adaptive* selection, and free of any dependence on $|\mathcal{C}|$. We report two versions: SPLIT uses a t-interval on the held-out half, which is the like-for-like comparison with the (also asymptotic) bootstrap band, and SPLIT-EB uses the finite-sample bound of (a) at $m = 1$. Comparing half-widths of the two asymptotic procedures, the band is narrower than a 50/50 split exactly when $\kappa \le \sqrt{2}$, i.e. when $m_{\text{eff}} \lesssim 5$ at $\alpha = 0.05$. That comparison is not the whole story, because splitting also degrades the *selection*: $\hat{c}$ is chosen from half the instances, so its true faithfulness is lower. The net effect has to be measured, and it goes the band's way whenever no hypothesis is clearly best and splitting's way when one is (Table 3).

(e) Selective intervals for the winner (cond, hybrid). Conditioning on the event $\{\arg\max = \hat{c}\}$ makes $\hat{\theta}_n(\hat{c})$ a truncated normal [2, 33], which can be inverted for a conditionally valid bound. We implement it, and report that it is informative only when the winner wins clearly; with several near-tied hypotheses — the normal situation in circuit search — the truncation point approaches the observed value and the interval degenerates. The hybrid construction of Andrews et al. [2] repairs this; ours is a conservative variant of it (we intersect with a two-sided simultaneous band rather than the exact projection region, so the level is at least $1 - \alpha$ and in practice somewhat above). We include both for completeness and recommend BOOT-MAX as the default.

Remark 1 (Randomised selection is not a free lunch)**.** For a Gaussian mean vector one can select on $U = \hat{\theta} + \tau\xi$ and infer from $V = \hat{\theta} - \tau^{-1}\hat{\Sigma}\xi/n$, which are independent [22, 34]. Setting $\tau^2 = a\sigma^2/n$ gives $\text{Var}(U) = \sigma^2(1+a)/n$ and $\text{Var}(V) = \sigma^2(1+1/a)/n$ — exactly the variances of an $f = 1/(1+a)$ sample split. Randomisation buys a continuous knob, not extra information; recovering information requires data carving [21], which we leave to future work.

4.1 Metrics that are not means

The most widely reported faithfulness metric is the *normalised logit difference recovered*,

$$F(c) = \frac{\mathbb{E}[\text{LD}(c)] - \mathbb{E}[\text{LD}(\emptyset)]}{\mathbb{E}[\text{LD}(\text{full})] - \mathbb{E}[\text{LD}(\emptyset)]},$$

a smooth functional of a mean vector rather than a mean. Its influence function is given in Appendix B; feeding it to the same bootstrap-t gives a simultaneous band for the whole family.

The finite-sample bounds (a)–(b) do not apply directly because F is not an average of bounded quantities; we give a conservative Fieller-style alternative in the appendix.

4.2 Adaptive search

A greedy pruner issues thousands of queries, each depending on the previous answers, so a union bound over the queries *actually asked* is not valid — the query set is itself random. Three options remain. **(i) Split.** Search on one part, certify on a held-out part. In interpretability the evaluation data is usually synthetic and nearly free, which makes this the default recommendation. **(ii) Uniformity over the reachable set.** A union or Occam bound over every circuit the search could have returned. For greedy removal of at most d of N components, $\log|\text{reachable}| = \log\sum_{k\leq d}\binom{N}{k}$, which is 99.8 nats for $N = 144, d = 124$ — a Hoeffding penalty of 0.158 at $n = 2000$. **(iii) Reusable holdout.** Thresholdout [18] answers adaptive queries from a holdout through a noisy threshold, spending budget only on queries that overfit. Its theoretical constants are loose, so we report it as an empirically validated option rather than a certificate.

5 What population? Clustered interventions

Interpretability datasets are generated from templates: the IOI dataset is 15 sentence frames in two name orders, filled with random names, objects and places. Resampling *prompts* and resampling *templates* answer different questions:

$$\theta_{\text{fix}}: \text{ new fills of these templates,} \qquad \theta_{\text{new}}: \text{ new templates as well.}$$

With G templates, n/G prompts each and intra-template correlation ρ_I, the variance of $\hat{\theta}$ under template-level sampling is inflated by Kish's design effect $\text{DEFF} = 1 + (n/G - 1)\rho_I$ [31], so the effective sample size is n/DEFF and saturates at G/ρ_I however many prompts are generated. For the simultaneous band we replace the i.i.d. resampling by the wild cluster bootstrap [10, 20] with Rademacher multipliers at the template level, studentised by the cluster-robust standard error, and inflate the critical value by t_{G-1}/z when G is small [9]. With G clusters the Rademacher weights admit only 2^G distinct sign assignments, so the attainable critical values are discrete at small G [36, 50]. Neither estimand is the "right" one — but a paper should say which it means, and we recommend reporting both.

6 Experiments

Design that removes oracle noise. For every setting we compute a per-instance score matrix once over a large pool of instances, then *treat the pool as the population*: analysis samples of size n are drawn i.i.d. from it. The true faithfulness $\theta(c)$ of every hypothesis is then the pooled column mean, known exactly. A measured coverage is therefore a Monte-Carlo average of an event whose truth value we know, and its only error is Monte-Carlo error over replications (with $R = 400$ replications the standard error near 0.95 is 0.011); it is not contaminated by uncertainty about the truth, which is the usual obstacle to measuring coverage on a real model. All bounds are one-sided at $\alpha = 0.05$; coverage means $\mathbb{P}(L(\hat{c}) \leq \theta(\hat{c}))$, and among procedures that attain it we rank by the *certified value* $L(\hat{c})$ — higher is better.

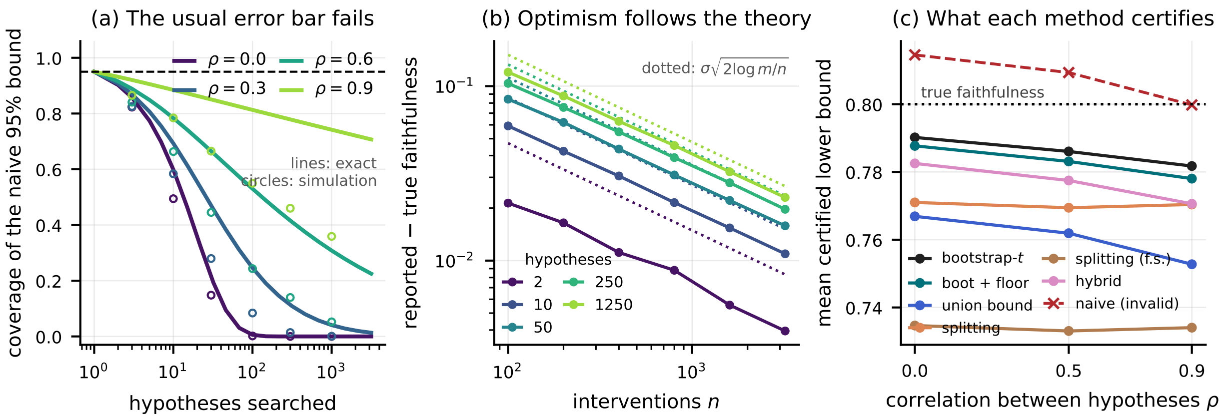

Figure 1: The anatomy of selection optimism, in a controlled setting where every hypothesis is equally faithful ($\theta(c) \equiv 0.80$). *(a)* Coverage of the ordinary "score $\pm z{\cdot}$SE" lower bound at the best-scoring hypothesis, as a function of how many hypotheses were searched. Lines are the exact expression of Proposition 3; circles are simulations with binary scores at $n = 500$. The interval keeps its nominal 95% only when a single hypothesis was considered. Correlation between hypotheses (ρ) is what stops the collapse from being total. *(b)* The inflation of the reported score follows the predicted $\sigma\sqrt{2\log|\mathcal{C}|/n}$ law (dotted) closely over two orders of magnitude in n. *(c)* At $|\mathcal{C}| = 200$, $\rho = 0.5$, $n = 1000$: every method except the naive one attains nominal coverage, and among those the bootstrap-t band certifies the highest lower bound. The conditional (selective) interval is valid but uninformative here, because many hypotheses are nearly tied.

6.1 E1: controlled simulations

Figure 1 isolates the phenomenon in a setting where we control the class size m, the number of interventions n, and the correlation ρ between hypotheses, and where every hypothesis has the same true faithfulness so that any apparent winner is pure selection noise. Panel (a) confirms Proposition 3 numerically and shows that the exact Gaussian expression tracks simulated binary (interchange-accuracy-like) scores closely. Panel (b) confirms the $\sqrt{2\log m/n}$ law of Propositions 1 and 2. Panel (c), and Table 3 in the appendix, answer the practical question: all of the corrected methods hold their nominal level, and the ranking among them by *certified value* is governed by m_{eff}. When hypotheses are strongly correlated the bootstrap band is the clear winner; when the class is large and the hypotheses are independent, sample splitting overtakes it. The conditional interval is valid throughout but collapses to an uninformative bound whenever the winner does not win clearly, which under the complete null is almost always.

6.2 E2: tiny transformers with known algorithms

We train four small transformers to 100% accuracy on tasks whose high-level causal model is known by construction (Appendix C), and search over a pre-registered class of alignment or circuit hypotheses. The four settings were chosen to span the regimes that matter.

(a) Hierarchical equality, searched (TT-A). A two-layer transformer computing $\mathbf{1}\{(a{=}b) = (c{=}d)\}$. Searching 435 alignment hypotheses for the variable $V_1 = \mathbf{1}\{a{=}b\}$ finds a clear winner: the whole residual stream at the position of token b after layer 0, with interchange accuracy 0.997. This is the benign regime — one hypothesis is genuinely much better than the rest, and the correction costs almost nothing.

(b) The same search restricted to localised subspaces (TT-A-local). An investigator who wants a *low-dimensional* representation pre-registers only the subspaces of at most 16 dimensions. That

class has no dominant member, and its top hypotheses are nearly tied: the near-tie regime, where selection bites.

(c) Three-term modular arithmetic (TT-C). A three-layer transformer computing $(a+b+c)$ mod 10. Searching 464 alignments for the intermediate sum $S_1 = (a+b)$ mod 10 finds *nothing*: the best population interchange accuracy is 0.128 against a chance level of 0.10. The class is not at fault: training the same architecture with interchange intervention training so that S_1 *is* carried by a 32-dimensional subspace after layer 1 produces a model in which that very hypothesis — a member of the same pre-registered class — attains interchange accuracy 1.000 (Appendix C). So the class can express a faithful alignment when one exists; the ordinarily trained network simply does not have one at any single position. This is the regime that matters most, because it is where a naive report is most dangerous: at $n = 1000$ the best-scoring hypothesis is reported as 0.134 on average, which an unwary reader would take as evidence of partial localisation.

(d) Induction circuits (TT-B). An attention-only two-layer model on the induction task, with the hypothesis class the full power set of its 8 heads. Here the *unconstrained* arg-max is uninteresting — 75 of the 256 subsets reproduce the full model's predictions on every instance — so we report the selection an investigator would actually make, the best circuit of at most two heads, whose true faithfulness is 0.995. This setting also exercises the Occam bound with the size-stratified prior.

Table 1 and Figure 2 report the coverage. The pattern matches the theory: naive coverage degrades exactly where the class contains near-ties, and every corrected bound holds its level everywhere.

Does the correction cost us the right answer? A valid bound is worthless if it makes the search useless. We check this on a fifth model, TT-A-IIT: the same equality architecture, trained by interchange intervention training [24] so that a specific 16-dimensional subspace after layer 0 *is* a faithful carrier of V_1 by construction. The planted hypothesis attains population interchange accuracy 1.000, and the search returns a hypothesis with that same accuracy in 100% of replications at $n = 1000$. Selection is not the problem; reporting the selected score without a correction is.

6.3 E3: the IOI circuit in GPT-2 small

We reproduce the standard indirect-object-identification evaluation [49]: 6000 prompts from 30 templates, on which GPT-2 small ranks the indirect object above the subject 99.5% of the time with a mean logit difference of 3.64. Every attention head outside a circuit is replaced by its average activation over the "ABC" distribution — the same prompts with three distinct names, so that name information is removed while sentence structure is preserved — computed separately for each template and each token position. This node-level scheme is the one used by automated circuit discovery [16]; Wang et al. [49] additionally restrict each circuit head to a single token position, so the two schemes are not directly comparable and neither dominates. Under ours the published 26-head circuit recovers 88% of the model's logit difference, close to the 87% they report under theirs. The statistical machinery does not care which scheme is used — it consumes whatever per-instance score the investigator defines — but the reported number does, and a paper should say which one it means.

Our hypothesis class contains 840 circuits, built without looking at the data from the seven functional head groups of Wang et al. [49] and from size-stratified random subsets. Table 2 is the headline result. At $n = 1000$ interventions the naive interval covers 59% of the time; the bootstrap band covers 99% and gives up only 0.007 of faithfulness relative to the (inflated) point estimate. The per-instance scores of different circuits are correlated (median pairwise correlation 0.28), and searching inflates the interval by a factor of $\kappa = 3.72$ against a Bonferroni factor of 2.34 for 840

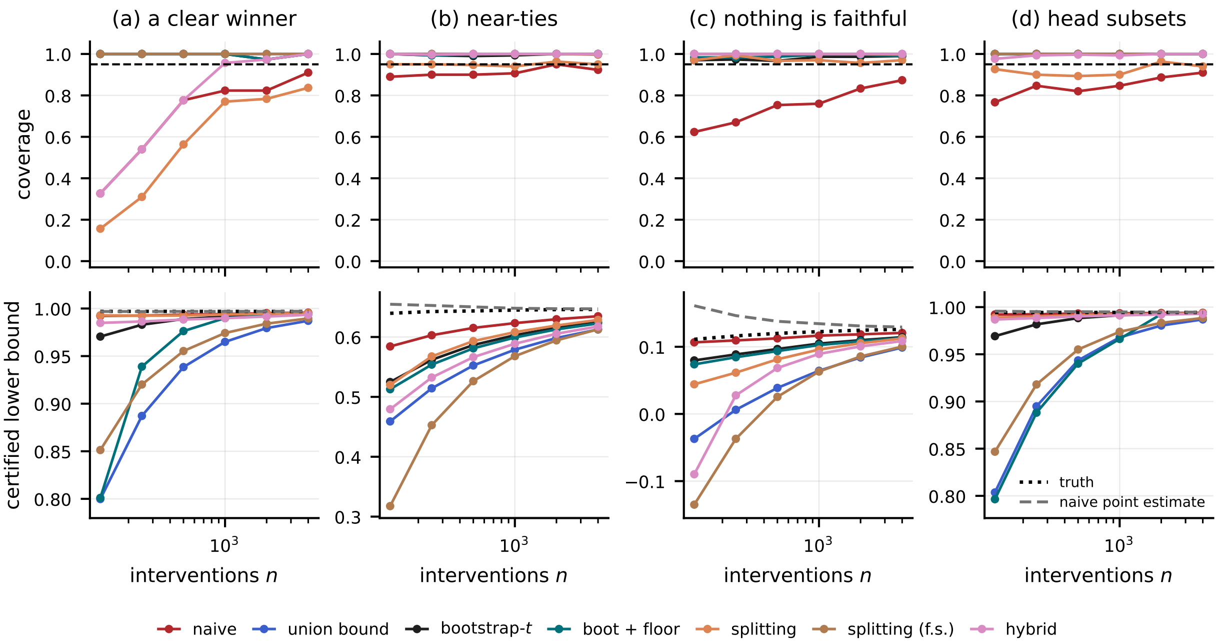

Figure 2: Coverage (top) and certified faithfulness (bottom) in four regimes on the tiny transformers. *(a)* equality, full alignment class: one hypothesis is far better than the rest. *(b)* the same task restricted to subspaces of at most 16 dimensions, where the top hypotheses are near-tied. *(c)* arithmetic, where no hypothesis in the class is faithful at all. *(d)* induction, selecting the best circuit of at most two attention heads. The horizontal axis is the number of interventions used for *both* search and inference; the dashed line is the nominal 95% level. In the bottom row the dotted black line is the true faithfulness of the selected hypothesis and the grey dashed line is the naive point estimate; the gap between them is the selection optimism. A valid method's curve must stay below the black dotted line. The conditional interval is omitted from the plot because it is off scale (it degenerates whenever hypotheses are near-tied); it appears in Table 1.

independent hypotheses. Here κ is the larger of the two, because a large fraction of the class is perfect on every sampled instance: the price being paid is not multiplicity but the heavy tail of a studentised maximum over near-degenerate columns, which is exactly what the floored variant addresses.

Certifying a whole frontier. Interpretability papers rarely report one circuit; they report a curve of faithfulness against circuit size and invite the reader to pick a point on it. Every point on that curve is a separate selection. Because our band is simultaneous, it certifies all of them at once: Figure 3(b) shows that the entire certified frontier is valid with probability 0.98, whereas the naive curve exceeds the truth at 0/43 of its points.

The unit of replication. Table 4 makes the second practical point. When prompts are drawn template-by-template — which is how an IOI evaluation set is actually built — the intra-template correlation of per-instance scores is 0.06, giving a design effect of 4.6 at $n = 2000$: those 2000 prompts carry about as much information as 433 independent ones. The naive interval then covers 73% of the time, and — the diagnostic point — its coverage *falls* as n grows (81% at $n = 500$), because more prompts from the same templates sharpen the search without sharpening the estimate. The i.i.d. band degrades the same way, from 100% at $n = 500$ to 95% at $n = 2000$; the cluster bootstrap holds its level throughout (100%). The effective sample size saturates at G/ρ_I, so the fix

Table 1: Exact coverage on tiny transformers with known algorithms. All four networks reach 100% task accuracy. Because the cached pool of 60,000 interventions is treated as the population, the true faithfulness of every hypothesis is known exactly and these coverages are exact, not estimated. The naive interval loses coverage wherever the class contains near-ties; the post-selection-valid bounds hold everywhere. Nominal level 0.95.

			coverage				certified $L(\hat{c})$		
setting	n	optimism	naive	boot	union	split	boot	union	split
TT-A equality, full class	250	$+-0.000$	0.54	1.00	1.00	0.31	0.983	0.887	0.992
	1000	$+-0.000$	0.82	1.00	1.00	0.77	0.991	0.965	0.994
	4000	+0.000	0.91	1.00	1.00	0.84	0.994	0.987	0.995
TT-A equality, localised	250	+0.010	0.90	0.99	1.00	0.95	0.562	0.514	0.568
	1000	+0.003	0.91	0.99	1.00	0.94	0.603	0.579	0.608
	4000	+0.001	0.92	1.00	1.00	0.95	0.625	0.612	0.629
TT-C arithmetic, null class	250	+0.030	0.67	0.97	1.00	0.99	0.089	0.006	0.061
	1000	+0.012	0.76	0.99	1.00	0.97	0.105	0.064	0.096
	4000	+0.003	0.87	0.99	1.00	0.97	0.114	0.099	0.113
TT-B induction, size ≤ 2	250	+0.001	0.85	1.00	1.00	0.90	0.982	0.895	0.990
	1000	+0.000	0.85	1.00	1.00	0.90	0.991	0.968	0.992
	4000	+0.000	0.91	1.00	1.00	0.94	0.993	0.987	0.993

is to write more templates, not to generate more prompts.

Pre-registration is the cheapest fix. The published IOI circuit was specified before we generated any data, so a single marginal interval is valid for it *with respect to our search.* At $n = 1000$ that interval certifies 0.978 — 0.009 more than what the same data can certify for a circuit chosen by searching our class. Fixing the hypothesis first and then generating fresh interventions removes the multiplicity penalty entirely. Two honest caveats. The circuit is only pre-registered relative to *us*: it was itself the product of a search by Wang et al. [49] over the same model and behaviour, and our interval does not correct for that search. And our hypothesis class is built around their circuit, so the comparison measures the cost of our search on top of theirs, not the cost of search in general.

6.4 E4: adaptive search and the reusable holdout

Everything so far assumes the hypothesis class was written down in advance. Here we let a greedy pruner choose instead, and ask which way of accounting for its queries still yields a usable number.

A greedy pruner asks the data thousands of adaptive questions, so neither a Bonferroni correction over the questions asked nor a simultaneous band over a pre-enumerated class applies. We measure what happens on a four-layer transformer with 36 ablatable components (attention heads and MLPs) trained to 100% accuracy on the induction task. The pruner removes components one at a time down to 8, issuing 631 queries against 250 interventions, and the whole procedure is repeated 40 times with the truth computed exactly on a pool of 40,000 interventions.

The first finding is a negative one, and worth stating plainly: on this model an adaptive search of 631 queries *barely overfits at all.* Reporting the search set's own score is biased by +0.0000; searching directly on a holdout and reporting the holdout score by +0.0000; answering through Thresholdout [18] by -0.0001. The reason is visible in the truth: the returned circuit's true faithfulness is 1.0000, because many small subsets of this network reproduce its predictions exactly. Adaptive search is not

Table 2: Certifying the selected circuit on GPT-2 small (IOI). The investigator picks the highest-scoring of $|\mathcal{C}| = 840$ circuit hypotheses using $n = 1000$ interventions and reports its faithfulness (behavioural agreement: does the ablated model still rank the indirect object above the subject?). Coverage is the probability that the reported lower bound really is below the selected circuit's true faithfulness; the nominal level is 0.95. "Certified" is the mean reported lower bound — among methods that achieve nominal coverage, higher is better. The naive interval, which ignores the search, covers 59% of the time. Over 350 independent replications; † marks coverage significantly below nominal.

method	coverage	certified $L(\hat{c})$
naive $\pm z$ SE *(invalid)*	0.589†	0.997
union bound, Hoeffding	1.000	0.929
union bound, empirical Bernstein	1.000	0.969
union bound, best of two	1.000	0.967
Occam bound (size prior)	1.000	0.814
bootstrap simultaneous band	0.986	0.992
bootstrap band, finite-sample floor	1.000	0.986
sample splitting (50/50)	0.794†	0.991
sample splitting, finite-sample	1.000	0.972
conditional (selective)	0.791†	-2.959
hybrid selective	0.791†	0.990

Selection diagnostics. Naive point estimate 0.998; true faithfulness of the selected circuit 0.997; optimism +0.002. Multiplicity factor $\kappa = 3.72$, against a Bonferroni factor of 2.34 for $|\mathcal{C}| = 840$ independent hypotheses.

automatically catastrophic, and a paper that holds data out is not thereby buying much accuracy.

What it *is* buying is a usable guarantee, and the price of the alternative is the second finding. Being uniform over every circuit the search could have returned means covering 11 orders of magnitude worth of subsets ($\log |\text{reachable}| = 25$ nats); the resulting bound is valid (100% coverage) but certifies only 0.732 against a truth of 1.0000. Holding out 250 interventions certifies 0.965 instead. The naive interval on the search data covers 92%. The same accounting holds on GPT-2 small (Figure 4, Table 5). A greedy pruner over a pre-registered pool of 60 candidate heads, using 200 interventions, issues 1,776 queries and returns a 11-head circuit. Its own search set says the circuit recovers 0.917 of the model's logit difference; 1200 held-out interventions say 0.916. The largest gap anywhere on the pruning path is 0.033. A bound uniform over every circuit the search could have reached ($\log |\text{reachable}| = 42$ nats) certifies 0.601; a held-out set of the same size as the search set already certifies 0.866.

The practical conclusion is blunt: *if the search is adaptive, hold data out.* It costs almost nothing, because interpretability evaluation sets are synthetic and nearly free to enlarge, whereas correcting after the fact costs almost everything.

7 A protocol for reporting faithfulness

Everything above reduces to a short checklist. It costs one extra column in a results table and no extra model forwards beyond the ones already being run.

1. **Write down the hypothesis class before you look at the data, and report its size.** "All subsets of these 26 heads", "all (layer, position, k-dimensional coordinate block) triples" — whatever it is. If you cannot name it, you cannot correct for it, and you should hold data out instead.

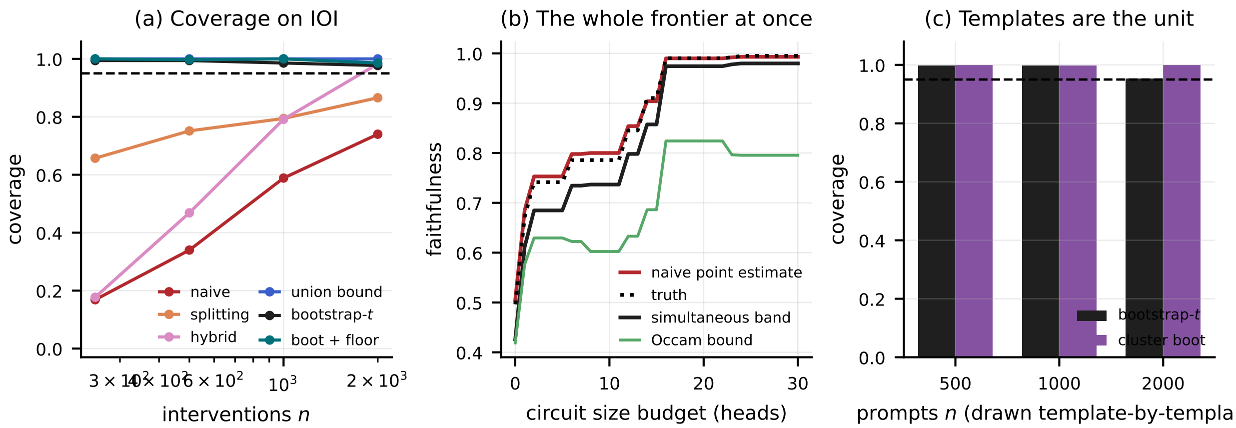

Figure 3: GPT-2 small, IOI. *(a)* Coverage of each bound at the selected circuit as the number of interventions grows. The naive interval does not improve with n — more data makes the search sharper as fast as it makes the estimate sharper. *(b)* The size/faithfulness frontier. For every size budget on the horizontal axis, the best-scoring circuit of at most that size is selected and its faithfulness reported. The naive curve (red) lies above the truth (dotted) at every budget; the simultaneous band (black) lies below it at every budget *jointly*, with probability 0.98. One band certifies the entire reported curve, not one point on it. *(c)* When prompts are drawn template-by-template — the honest model of how the evaluation set was produced — a bound that treats prompts as independent under-covers badly, and the cluster bootstrap restores validity.

2. **Say which population your interventions represent.** New fills of your templates, or new templates? The two give different error bars (Section 5); report the cluster-robust one if you mean the second, and say which you mean.

3. **Keep the per-instance scores, not just their average.** The $n \times |\mathcal{C}|$ matrix is what every bound in this paper consumes; it costs nothing extra to save and it lets a reader recompute any interval.

4. **Freeze the ablation values.** Mean ablation replaces a component by an average, and if that average is computed from the evaluation sample itself then the per-instance score is not a function of a single instance and none of the bounds apply as stated. Compute the ablation values once, on a reference set disjoint from the evaluation instances, and hold them fixed. (Resample ablation has the same issue and is not covered by our guarantees at all.)

5. **Pick a bound.** *(i)* If the class was fixed in advance and is enumerable, use the bootstrap-t simultaneous band — it is the tightest, it certifies every hypothesis you looked at, and it adapts to the correlation between them. If some hypotheses may be perfect on every sampled instance, use the floored variant. *(ii)* If the class is a power set of components, use the Occam bound with the size-stratified prior. *(iii)* If the search was adaptive (greedy pruning, gradient-based alignment search, or an investigator iterating), hold interventions out. In interpretability the evaluation data is synthetic and nearly free, so this is almost always the right answer.

6. **Report the certified lower bound next to the point estimate**, and report m_{eff}. A reader then sees both the number and its warranty.

7. **Pre-register the final circuit.** The cheapest way to get a tight interval is to fix the circuit, generate fresh interventions, and evaluate once: the multiplicity penalty disappears entirely. In our GPT-2 experiment this is worth 0.009 of certified faithfulness at $n = 1000$ (Table 2

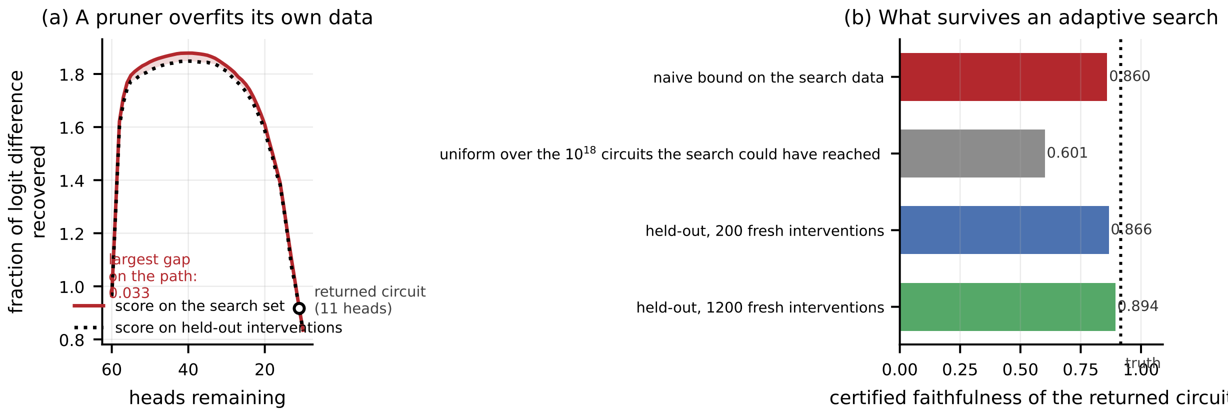

Figure 4: A greedy pruner overfits the interventions it prunes on (GPT-2 small, IOI). *(a)* As heads are removed one at a time, the fraction of the model's logit difference that the surviving circuit recovers, scored on the 200 interventions the pruner itself used (solid) and on 1200 held-out interventions (dotted). The gap between them is what a same-data report would hide. *(b)* What can be certified for the circuit the pruner returns. A bound that is uniform over every circuit the search could have reached is valid but uninformative at this sample size; a held-out set of the same size as the search set already certifies most of the truth (dotted line).

versus the pre-registered row in Appendix D).

8 Related work

Causal abstraction and circuit faithfulness. Geiger et al. [26] give the theoretical foundation we build on, unifying activation patching, causal scrubbing [11], distributed alignment search [25] and related methods; the graded notion of faithfulness we bound is theirs. Circuit analyses [28, 35, 38, 49] and automated discovery methods [6, 16, 32, 46] supply the searches whose selection effects we quantify. Zhang and Nanda [52] and Heimersheim and Nanda [29] give methodological guidance on activation patching, and Miller et al. [40] show that faithfulness metrics are sensitive to ablation choices; both are complementary to our question, which is about the sampling and selection uncertainty of whatever metric is chosen. Makelov et al. [37] exhibit interpretability illusions in subspace patching — a hypothesis can score well for the wrong reason — which is a validity concern orthogonal to, and not addressed by, confidence bounds.

Statistical testing in interpretability. Closest to us is Shi et al. [44], who test a *given* circuit hypothesis against nulls built from random or shuffled circuits. Their tests assume the circuit was fixed in advance; our contribution is precisely the selection step that precedes such a test. The field is not innocent of multiplicity either: Wang et al. [49] already check *completeness* and *minimality* by evaluating their circuit over a distribution of knockout subsets rather than at a single point, which is a multiplicity-aware check in spirit, and the distributed-alignment-search line [25, 51] trains its alignments on one split and evaluates on another, so the premise of this paper does not apply to it without qualification. What is missing across all of them is a confidence bound on the reported number that survives the search — which is what we supply. Adolfi et al. [1] show that circuit discovery is computationally intractable in the worst case, a complementary negative result about search rather than inference.

Post-selection and adaptive inference. Valid post-selection inference [5], exact selective inference after model selection [21, 33, 47, 48], inference on winners [2], post-selection inference via algorithmic stability [54] and splitting strategies [22, 34] form the statistical backbone. Adaptive data analysis [4, 7, 17–19, 43] provides the tools for sequential search. Our simultaneous bands rest on high-dimensional Gaussian approximation [12–14] and on empirical-Bernstein concentration [3, 39]. The phenomenon is familiar elsewhere: the winner's curse in genetics [53], adaptive overfitting on benchmarks [41, 42], and researcher degrees of freedom in the empirical sciences [27, 30, 45]. To our knowledge none of this machinery has been brought to bear on mechanistic faithfulness, and the specific structure of the problem — bounded per-instance scores, strongly overlapping hypotheses, template-clustered interventions, and metrics that are ratios of means — requires the specific combination we assemble.

9 Limitations

We bound sampling and selection error, not construct validity. A confidence bound answers "if I ran this same evaluation on new interventions, what would I get?". It does not answer "is interchange-intervention accuracy the right thing to measure?". A hypothesis can score highly for the wrong reason: subspace patching admits interpretability illusions [37], and faithfulness scores move when the ablation scheme changes [40, 52]. Our bounds inherit whatever the chosen metric means; they make the number reliable, not necessarily meaningful. The two concerns compose: an illusory hypothesis with a tight, valid bound is still illusory.

Estimated ablation values. Mean ablation replaces a component by an average over a reference distribution. In our experiments that average is computed once, before any resampling, and held fixed, so conditional on it the per-instance scores are i.i.d. functions of a single instance and the guarantees apply as stated. A practitioner who recomputes the ablation values inside each evaluation — or who uses resample ablation — is outside the stated guarantees, which is why the protocol asks for the values to be frozen.

Few clusters. Template-level inference is limited by the number of templates, not the number of prompts. With the 30 templates we wrote, the cluster-robust band covers 100% rather than the nominal 95%; our simulations put the shortfall at about three points for $G \approx 25$ and none for $G \geq 50$ (Table 4). Few-cluster inference is known to remain anti-conservative [36]; the honest reading is that a template-level guarantee needs more templates, not more prompts.

The bootstrap band is asymptotic. Its validity rests on Gaussian approximation for maxima [12, 15], which requires variances bounded away from zero — a condition that a hypothesis scoring exactly 1 on every sampled instance violates. We floor the studentising standard deviation at $n^{-1/2}$, provide the finite-sample-floored variant of Section 4(c) for classes where this matters, and verify coverage empirically in every setting we report. The empirical-Bernstein and Occam bounds have no such caveat but pay a $\log |\mathcal{C}|$ price.

Continuous hypothesis classes. Distributed alignment search optimises a rotation of the residual stream by gradient descent [25], so its hypothesis class is continuous and its search is adaptive. We handle only finite, pre-enumerated classes directly; for the continuous case the honest options are sample splitting (which we recommend) or an Occam bound over a discretisation, which requires a covering-number argument we do not develop here.

Our hypothesis classes are ours, not the field's. The magnitude of the optimism we report on GPT-2 depends on the class we enumerate ($|\mathcal{C}| = 840$ circuits built from the seven functional head groups of Wang et al. [49] plus size-stratified random subsets). A different class would give different numbers. That dependence is the point — optimism is a property of the search, not of the model — but it means our specific figures should be read as an illustration of magnitude rather than as a universal constant.

The template population is a modelling choice. Our cluster-robust analysis resamples the 30 templates we wrote. Those templates are themselves a sample from an ill-defined population of "sentences that pose an indirect-object-identification problem", and no resampling scheme can capture that. We report the cluster-robust interval as a lower bound on the honest uncertainty about the behaviour, not as its exact quantification.

Reusable-holdout constants are loose. Thresholdout [18] comes with a generalisation guarantee whose constants are far too conservative to yield a usable certificate at our sample sizes. We therefore report it as an empirically validated mechanism, and recommend a plain held-out set when a guarantee is required.

Scale. All experiments run on four tiny transformers and GPT-2 small, within 5.1 GPU-hours on a single consumer accelerator. Nothing in the statistics depends on model size — the input is a score matrix — but we have not verified that the empirical magnitudes (effective number of hypotheses, intra-template correlation) transfer to larger models, where circuits are less sparse and behaviours more heterogeneous.

10 Conclusion

Causal abstraction made circuit hypotheses falsifiable. Falsifiability is a property of a hypothesis fixed in advance; a circuit that was chosen by maximising the same score it is then evaluated on has not been tested in that sense. The correction is well understood in statistics, it is cheap in interpretability's usual regime, and it converts a point estimate into a claim a reader can rely on. We have supplied the estimator, the guarantees, the validation, and the code.

References

[1] Federico Adolfi, Martina G. Vilas, and Todd Wareham. The computational complexity of circuit discovery for inner interpretability. In *International Conference on Learning Representations (ICLR)*, 2025. URL https://openreview.net/forum?id=QogcGNXJVw. Spotlight; arXiv:2410.08025.

[2] Isaiah Andrews, Toru Kitagawa, and Adam McCloskey. Inference on winners. *The Quarterly Journal of Economics*, 139(1):305–358, 2024. doi: 10.1093/qje/qjad043.

[3] Jean-Yves Audibert, Rémi Munos, and Csaba Szepesvári. Exploration–exploitation tradeoff using variance estimates in multi-armed bandits. *Theoretical Computer Science*, 410(19): 1876–1902, 2009. doi: 10.1016/j.tcs.2009.01.016.

[4] Raef Bassily, Kobbi Nissim, Adam Smith, Thomas Steinke, Uri Stemmer, and Jonathan Ullman. Algorithmic stability for adaptive data analysis. In *Proceedings of the Forty-Eighth Annual*

ACM SIGACT Symposium on Theory of Computing (STOC), pages 1046–1059. ACM, 2016. doi: 10.1145/2897518.2897566.

[5] Richard Berk, Lawrence Brown, Andreas Buja, Kai Zhang, and Linda Zhao. Valid post-selection inference. *The Annals of Statistics*, 41(2):802–837, 2013. doi: 10.1214/12-AOS1077. URL https://doi.org/10.1214/12-AOS1077.

[6] Adithya Bhaskar, Alexander Wettig, Dan Friedman, and Danqi Chen. Finding Transformer circuits with edge pruning. In A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang, editors, *Advances in Neural Information Processing Systems*, volume 37, pages 18506–18534. Curran Associates, Inc., 2024. doi: 10.52202/079017-0587. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/20fdaf67581e6d7157376d1ed584040a-Abstract-Conference.html. Spotlight. arXiv:2406.16778.

[7] Avrim Blum and Moritz Hardt. The Ladder: A reliable leaderboard for machine learning competitions. In *Proceedings of the 32nd International Conference on Machine Learning (ICML)*, volume 37 of *Proceedings of Machine Learning Research*, pages 1006–1014, Lille, France, 2015. PMLR.

[8] Stéphane Boucheron, Gábor Lugosi, and Pascal Massart. *Concentration Inequalities: A Nonasymptotic Theory of Independence*. Oxford University Press, 2013. ISBN 9780199535255. doi: 10.1093/acprof:oso/9780199535255.001.0001. URL https://doi.org/10.1093/acprof:oso/9780199535255.001.0001.

[9] A. Colin Cameron and Douglas L. Miller. A Practitioner's Guide to Cluster-Robust Inference. *Journal of Human Resources*, 50(2):317–372, 2015. ISSN 0022-166X. doi: 10.3368/jhr.50.2.317. URL https://doi.org/10.3368/jhr.50.2.317.

[10] A. Colin Cameron, Jonah B. Gelbach, and Douglas L. Miller. Bootstrap-based improvements for inference with clustered errors. *The Review of Economics and Statistics*, 90(3):414–427, 2008. doi: 10.1162/rest.90.3.414.

[11] Lawrence Chan, Adrià Garriga-Alonso, Nicholas Goldowsky-Dill, Ryan Greenblatt, Jenny Nitishinskaya, Ansh Radhakrishnan, Buck Shlegeris, and Nate Thomas. Causal scrubbing: a method for rigorously testing interpretability hypotheses. AI Alignment Forum, Redwood Research, December 2022. URL https://www.alignmentforum.org/posts/JvZhhzycHu2Yd57RN/causal-scrubbing-a-method-for-rigorously-testing. Posted 3 December 2022; authors listed alphabetically.

[12] Victor Chernozhukov, Denis Chetverikov, and Kengo Kato. Gaussian approximations and multiplier bootstrap for maxima of sums of high-dimensional random vectors. *The Annals of Statistics*, 41(6):2786–2819, 2013. doi: 10.1214/13-AOS1161. arXiv:1212.6906.

[13] Victor Chernozhukov, Denis Chetverikov, and Kengo Kato. Empirical and multiplier bootstraps for suprema of empirical processes of increasing complexity, and related Gaussian couplings. *Stochastic Processes and their Applications*, 126(12):3632–3651, 2016. doi: 10.1016/j.spa.2016.04.009.

[14] Victor Chernozhukov, Denis Chetverikov, and Kengo Kato. Central limit theorems and bootstrap in high dimensions. *The Annals of Probability*, 45(4):2309–2352, 2017. doi: 10.1214/16-AOP1113. arXiv:1412.3661.

[15] Victor Chernozhukov, Denis Chetverikov, and Kengo Kato. Inference on Causal and Structural Parameters using Many Moment Inequalities. *The Review of Economic Studies*, 86(5):1867–1900, 2019. doi: 10.1093/restud/rdy065. URL https://doi.org/10.1093/restud/rdy065. Earlier circulated as "Testing many moment inequalities", arXiv:1312.7614.

[16] Arthur Conmy, Augustine N. Mavor-Parker, Aengus Lynch, Stefan Heimersheim, and Adrià Garriga-Alonso. Towards automated circuit discovery for mechanistic interpretability. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, volume 36, pages 16318–16352. Curran Associates, Inc., 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/34e1dbe95d34d7ebaf99b9bcaeb5b2be-Abstract-Conference.html. arXiv:2304.14997.

[17] Cynthia Dwork, Vitaly Feldman, Moritz Hardt, Toniann Pitassi, Omer Reingold, and Aaron Roth. Preserving statistical validity in adaptive data analysis. In *Proceedings of the Forty-Seventh Annual ACM Symposium on Theory of Computing (STOC)*, pages 117–126. ACM, 2015. doi: 10.1145/2746539.2746580.

[18] Cynthia Dwork, Vitaly Feldman, Moritz Hardt, Toniann Pitassi, Omer Reingold, and Aaron Roth. The reusable holdout: Preserving validity in adaptive data analysis. *Science*, 349(6248): 636–638, 2015. doi: 10.1126/science.aaa9375.

[19] Vitaly Feldman and Thomas Steinke. Generalization for adaptively-chosen estimators via stable median. In *Proceedings of the 2017 Conference on Learning Theory (COLT)*, volume 65 of *Proceedings of Machine Learning Research*, pages 728–757. PMLR, 2017.

[20] C. A. Field and A. H. Welsh. Bootstrapping clustered data. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 69(3):369–390, 2007. doi: 10.1111/j.1467-9868.2007.00593.x.

[21] William Fithian, Dennis Sun, and Jonathan Taylor. Optimal Inference After Model Selection. *arXiv preprint arXiv:1410.2597*, 2014. URL https://arxiv.org/abs/1410.2597.

[22] Daniel García Rasines and G. Alastair Young. Splitting strategies for post-selection inference. *Biometrika*, 110(3):597–614, 2023. doi: 10.1093/biomet/asac070. URL https://doi.org/10.1093/biomet/asac070. arXiv:2102.02159.

[23] Atticus Geiger, Hanson Lu, Thomas Icard, and Christopher Potts. Causal abstractions of neural networks. In M. Ranzato, A. Beygelzimer, Y. Dauphin, P. S. Liang, and J. Wortman Vaughan, editors, *Advances in Neural Information Processing Systems*, volume 34, pages 9574–9586. Curran Associates, Inc., 2021. URL https://proceedings.neurips.cc/paper_files/paper/2021/file/4f5c422f4d49a5a807eda27434231040-Paper.pdf.

[24] Atticus Geiger, Zhengxuan Wu, Hanson Lu, Josh Rozner, Elisa Kreiss, Thomas Icard, Noah Goodman, and Christopher Potts. Inducing causal structure for interpretable neural networks. In Kamalika Chaudhuri, Stefanie Jegelka, Le Song, Csaba Szepesvari, Gang Niu, and Sivan Sabato, editors, *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 7324–7338. PMLR, 2022. URL https://proceedings.mlr.press/v162/geiger22a.html.

[25] Atticus Geiger, Zhengxuan Wu, Christopher Potts, Thomas Icard, and Noah Goodman. Finding alignments between interpretable causal variables and distributed neural representations. In

Francesco Locatello and Vanessa Didelez, editors, *Proceedings of the Third Conference on Causal Learning and Reasoning*, volume 236 of *Proceedings of Machine Learning Research*, pages 160–187. PMLR, 2024. URL https://proceedings.mlr.press/v236/geiger24a.html.

[26] Atticus Geiger, Duligur Ibeling, Amir Zur, Maheep Chaudhary, Sonakshi Chauhan, Jing Huang, Aryaman Arora, Zhengxuan Wu, Noah Goodman, Christopher Potts, and Thomas Icard. Causal abstraction: A theoretical foundation for mechanistic interpretability. *Journal of Machine Learning Research*, 26(83):1–64, 2025. URL https://jmlr.org/papers/v26/23-0058.html.

[27] Andrew Gelman and Eric Loken. The garden of forking paths: Why multiple comparisons can be a problem, even when there is no "fishing expedition" or "p-hacking" and the research hypothesis was posited ahead of time. Unpublished manuscript, Department of Statistics, Columbia University, November 2013. URL https://sites.stat.columbia.edu/gelman/research/unpublished/p_hacking.pdf.

[28] Michael Hanna, Ollie Liu, and Alexandre Variengien. How does GPT-2 compute greater-than?: Interpreting mathematical abilities in a pre-trained language model. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, editors, *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, volume 36, pages 76033–76060. Curran Associates, Inc., 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/efbba7719cc5172d175240f24be11280-Abstract-Conference.html.

[29] Stefan Heimersheim and Neel Nanda. How to use and interpret activation patching. *arXiv preprint arXiv:2404.15255*, 2024. URL https://arxiv.org/abs/2404.15255.

[30] John P. A. Ioannidis. Why most published research findings are false. *PLoS Medicine*, 2(8): e124, 2005. doi: 10.1371/journal.pmed.0020124. URL https://doi.org/10.1371/journal.pmed.0020124.

[31] Leslie Kish. *Survey Sampling*. John Wiley & Sons, New York, 1965. OCLC 256017; xvi + 643 pp.; reissued 1995 as a Wiley Classics Library edition, ISBN 978-0-471-10949-5.

[32] János Kramár, Tom Lieberum, Rohin Shah, and Neel Nanda. AtP*: An efficient and scalable method for localizing LLM behaviour to components, 2024. URL https://arxiv.org/abs/2403.00745.

[33] Jason D. Lee, Dennis L. Sun, Yuekai Sun, and Jonathan E. Taylor. Exact post-selection inference, with application to the lasso. *The Annals of Statistics*, 44(3):907–927, 2016. doi: 10.1214/15-AOS1371. URL https://doi.org/10.1214/15-AOS1371. arXiv:1311.6238.

[34] James Leiner, Boyan Duan, Larry Wasserman, and Aaditya Ramdas. Data fission: Splitting a single data point. *Journal of the American Statistical Association*, 120(549):135–146, 2025. doi: 10.1080/01621459.2023.2270748. arXiv:2112.11079.

[35] Tom Lieberum, Matthew Rahtz, János Kramár, Neel Nanda, Geoffrey Irving, Rohin Shah, and Vladimir Mikulik. Does circuit analysis interpretability scale? Evidence from multiple choice capabilities in Chinchilla. *arXiv preprint arXiv:2307.09458*, 2023. URL https://arxiv.org/abs/2307.09458.

[36] James G. MacKinnon, Morten Ørregaard Nielsen, and Matthew D. Webb. Cluster-robust inference: A guide to empirical practice. *Journal of Econometrics*, 232(2):272–299, 2023. doi: 10.1016/j.jeconom.2022.04.001. URL https://doi.org/10.1016/j.jeconom.2022.04.001. arXiv:2205.03285.

[37] Aleksandar Makelov, Georg Lange, Atticus Geiger, and Neel Nanda. Is this the subspace you are looking for? an interpretability illusion for subspace activation patching. In B. Kim, Y. Yue, S. Chaudhuri, K. Fragkiadaki, M. Khan, and Y. Sun, editors, *The Twelfth International Conference on Learning Representations (ICLR)*, pages 26486–26515, 2024. URL https://proceedings.iclr.cc/paper_files/paper/2024/hash/70b8505ac79e3e131756f793cd80eb8d-Abstract-Conference.html. arXiv:2311.17030.

[38] Samuel Marks, Can Rager, Eric J. Michaud, Yonatan Belinkov, David Bau, and Aaron Mueller. Sparse feature circuits: Discovering and editing interpretable causal graphs in language models. In *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025. URL https://openreview.net/forum?id=I4e82CIDxv.

[39] Andreas Maurer and Massimiliano Pontil. Empirical Bernstein bounds and sample variance penalization. In *Proceedings of the 22nd Annual Conference on Learning Theory (COLT)*, 2009. URL https://arxiv.org/abs/0907.3740.

[40] Joseph Miller, Bilal Chughtai, and William Saunders. Transformer circuit faithfulness metrics are not robust. In *First Conference on Language Modeling (COLM)*, 2024. URL https://openreview.net/forum?id=zSf8PJyQb2. Spotlight. arXiv:2407.08734.

[41] Benjamin Recht, Rebecca Roelofs, Ludwig Schmidt, and Vaishaal Shankar. Do ImageNet classifiers generalize to ImageNet? In Kamalika Chaudhuri and Ruslan Salakhutdinov, editors, *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 5389–5400. PMLR, 2019. URL https://proceedings.mlr.press/v97/recht19a.html. arXiv:1902.10811.

[42] Rebecca Roelofs, Vaishaal Shankar, Benjamin Recht, Sara Fridovich-Keil, Moritz Hardt, John Miller, and Ludwig Schmidt. A meta-analysis of overfitting in machine learning. In *Advances in Neural Information Processing Systems 32 (NeurIPS 2019)*, pages 9175–9185, 2019. URL https://proceedings.neurips.cc/paper/2019/hash/ee39e503b6bedf0c98c388b7e8589aca-Abstract.html.

[43] Daniel Russo and James Zou. Controlling bias in adaptive data analysis using information theory. In *Proceedings of the 19th International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 51 of *Proceedings of Machine Learning Research*, pages 1232–1240, Cadiz, Spain, 2016. PMLR.

[44] Claudia Shi, Nicolas Beltran-Velez, Achille Nazaret, Carolina Zheng, Adrià Garriga-Alonso, Andrew Jesson, Maggie Makar, and David M. Blei. Hypothesis testing the circuit hypothesis in LLMs. In A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang, editors, *Advances in Neural Information Processing Systems*, volume 37, pages 94539–94567. Curran Associates, Inc., 2024. doi: 10.52202/079017-2997. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/abccb8a90b30d45b948360ba41f5a20f-Abstract-Conference.html. arXiv:2410.13032.

[45] Joseph P. Simmons, Leif D. Nelson, and Uri Simonsohn. False-positive psychology: Undisclosed flexibility in data collection and analysis allows presenting anything as significant. *Psychological Science*, 22(11):1359–1366, 2011. doi: 10.1177/0956797611417632. URL https://doi.org/10.1177/0956797611417632.

[46] Aaquib Syed, Can Rager, and Arthur Conmy. Attribution patching outperforms automated circuit discovery. In Yonatan Belinkov, Najoung Kim, Jaap Jumelet, Hosein Mohebbi, Aaron Mueller, and Hanjie Chen, editors, *Proceedings of the 7th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*, pages 407–416, Miami, Florida, US, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.blackboxnlp-1.25. URL https://aclanthology.org/2024.blackboxnlp-1.25/. arXiv:2310.10348.

[47] Xiaoying Tian and Jonathan Taylor. Selective inference with a randomized response. *The Annals of Statistics*, 46(2):679–710, 2018. doi: 10.1214/17-AOS1564. arXiv:1507.06739.

[48] Ryan J. Tibshirani, Jonathan Taylor, Richard Lockhart, and Robert Tibshirani. Exact Post-Selection Inference for Sequential Regression Procedures. *Journal of the American Statistical Association*, 111(514):600–620, 2016. doi: 10.1080/01621459.2015.1108848. URL https://doi.org/10.1080/01621459.2015.1108848. arXiv:1401.3889.

[49] Kevin Ro Wang, Alexandre Variengien, Arthur Conmy, Buck Shlegeris, and Jacob Steinhardt. Interpretability in the wild: a circuit for indirect object identification in GPT-2 Small. In *The Eleventh International Conference on Learning Representations (ICLR 2023)*. OpenReview.net, 2023. URL https://openreview.net/forum?id=NpsVSN6o4ul. arXiv:2211.00593.

[50] Matthew D. Webb. Reworking wild bootstrap-based inference for clustered errors. *Canadian Journal of Economics/Revue canadienne d'économique*, 56(3):839–858, 2023. doi: 10.1111/caje.12661. URL https://doi.org/10.1111/caje.12661.

[51] Zhengxuan Wu, Atticus Geiger, Thomas Icard, Christopher Potts, and Noah Goodman. Interpretability at scale: Identifying causal mechanisms in Alpaca. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, editors, *Advances in Neural Information Processing Systems*, volume 36, pages 78205–78226. Curran Associates, Inc., 2023. doi: 10.52202/075280-3418. URL https://proceedings.neurips.cc/paper_files/paper/2023/file/f6a8b109d4d4fd64c75e94aaf85d9697-Paper-Conference.pdf.

[52] Fred Zhang and Neel Nanda. Towards best practices of activation patching in language models: Metrics and methods. In B. Kim, Y. Yue, S. Chaudhuri, K. Fragkiadaki, M. Khan, and Y. Sun, editors, *The Twelfth International Conference on Learning Representations (ICLR)*, pages 1651–1678, 2024. URL https://proceedings.iclr.cc/paper_files/paper/2024/hash/06a52a54c8ee03cd86771136bc91eb1f-Abstract-Conference.html. arXiv:2309.16042.

[53] Sebastian Zöllner and Jonathan K. Pritchard. Overcoming the winner's curse: Estimating penetrance parameters from case-control data. *American Journal of Human Genetics*, 80(4):605–615, 2007. doi: 10.1086/512821.

[54] Tijana Zrnic and Michael I. Jordan. Post-selection inference via algorithmic stability. *The Annals of Statistics*, 51(4):1666–1691, 2023. doi: 10.1214/23-AOS2303. URL https://doi.org/10.1214/23-AOS2303. arXiv:2011.09462.

A Proofs

A.1 Proposition 1

Let $D_c = \hat{\theta}_n(c) - \theta(c)$ and $X_i = \varphi(c; Z_i) - \theta(c)$, so that $D_c = n^{-1} \sum_i X_i$, $\mathbb{E} X_i = 0$, $|X_i| \le 1$ and $\operatorname{Var} X_i \le \sigma^2_{\max}$. Bernstein's moment-generating-function bound for a centred variable with $|X| \le b$

gives $\log \mathbb{E}e^{\lambda X} \le \lambda^2 \operatorname{Var}(X)/(2(1-\lambda b/3))$ for $0 < \lambda < 3/b$; summing over n independent copies and rescaling by $1/n$,

$$\log \mathbb{E}e^{\lambda D_c} \;\le\; \frac{\lambda^2 v}{2(1-\kappa\lambda)}, \qquad v = \frac{\sigma_{\max}^2}{n}, \quad \kappa = \frac{1}{3n},$$

which is exactly the statement that D_c is *sub-gamma on the right tail* with variance factor v and scale parameter κ, in the sense of Boucheron et al. [8, §2.4]. Their maximal inequality [8, Cor. 2.6, p. 33] — which requires the variables to be centred but *not* independent of one another, so the strong correlation between circuit hypotheses is harmless — then gives

$$\mathbb{E}\max_{c\in\mathcal{C}} D_c \;\le\; \sqrt{2v\log m} + \kappa\log m \;=\; \sigma_{\max}\sqrt{\frac{2\log m}{n}} + \frac{\log m}{3n}.$$

Finally $\hat\theta_n(\hat c) - \theta(\hat c) \le \max_c D_c$ pointwise for any selection rule, so the same bound holds for its expectation. ☐

A.2 Proposition 2

Under the stated model $\sqrt{n}(\hat\theta_n(c)-\theta_0)/\sigma = N_c$ with $N_1,\dots,N_m$ i.i.d. standard normal, and the argmax selects the largest N_c, so $\hat\theta_n(\hat c) - \theta(\hat c) = (\sigma/\sqrt{n})\max_c N_c$ exactly. For i.i.d. standard normals, $\mathbb{E}\max_{j\le m} N_j \ge \frac{1}{\sqrt{2}}\sqrt{\log m}$ [8, Exercise 13.6], used there in the proof of Sudakov's minoration; combined with the sub-Gaussian upper bound $\mathbb{E}\max_{j\le m} N_j \le \sqrt{2\log m}$ this gives the claim. ☐

Remark 2 (The restriction to $m \ge 3$)**.** The bound with constant $1/\sqrt{2}$ fails at $m = 2$, where $\mathbb{E}\max_{j\le 2} N_j = 1/\sqrt{\pi} = 0.5642$ exactly while $\frac{1}{\sqrt{2}}\sqrt{\log 2} = 0.5887$. The ratio $\mathbb{E}[\max_{j\le m} N_j]/\sqrt{\log m}$ is minimised at $m = 2$ and increases thereafter, so the largest constant valid for every $m \ge 2$ is $1/\sqrt{\pi\log 2} \approx 0.6777$; we computed the ratio numerically up to $m = 10^5$ (0.678 at $m = 2$, 0.807 at $m = 3$, 0.874 at $m = 4$, 1.269 at $m = 10^4$). Nothing in the paper depends on the case $m = 2$.

A.3 Proposition 3

The naive bound at $\hat c$ is $\hat\theta_n(\hat c) - z_{1-\alpha}\hat\sigma(\hat c)/\sqrt{n}$, and (treating $\hat\sigma$ as σ, which is exact in the Gaussian limit) it covers iff $G_{\hat c} \le z_{1-\alpha}$. Since $\hat c = \arg\max_c \hat\theta_n(c)$ and all $\theta(c)$ and $\sigma(c)$ are equal, $\hat c = \arg\max_c G_c$, so the coverage event is exactly $\{\max_c G_c \le z_{1-\alpha}\}$. Writing the equicorrelated Gaussian vector as $G_c = \sqrt{\rho}\,W + \sqrt{1-\rho}\,\varepsilon_c$ with $W, \varepsilon_1, \dots, \varepsilon_m$ i.i.d. standard normal and conditioning on W makes the coordinates independent, giving the stated integral. ☐

A.4 Validity of the bounds in Section 4

(a) and (b). Both are unions of the Maurer–Pontil bound [39, Thm. 4], applied at level α/m and $\alpha\pi(c)$ respectively. In case (b), $\sum_c \alpha\pi(c) \le \alpha$ because π is a sub-probability, so the union bound applies. Both statements are *simultaneous* over $\mathcal{C}$, hence hold at the (random) $\hat c$:

$$\mathbb{P}\big(\theta(\hat c) < L(\hat c)\big) \le \mathbb{P}\big(\exists c : \theta(c) < L(c)\big) \le \alpha.$$

Crucially, this argument uses nothing about how $\hat c$ was chosen, so it covers greedy search, gradient-based alignment search, and human inspection alike.

(c). Write $x_{ic} = \varphi(c; Z_i) - \theta(c)$. For the *non-studentised* maximum $\max_c n^{-1/2} \sum_i x_{ic}$, Chernozhukov et al. [12, Thm. 3.2 and Cor. 3.1] show that the Gaussian multiplier bootstrap quantile satisfies $\sup_{\alpha \in (0,1)} |\mathbb{P}(T \leq c_W(\alpha)) - \alpha| \leq Cn^{-c}$ provided (i) $c_1 \leq n^{-1} \sum_i \mathbb{E}[x_{ic}^2] \leq C_1$ for every c and (ii) $B_n^2(\log(mn))^7/n \leq C_2 n^{-c_2}$, where B_n is an envelope constant. Our scores are bounded in $[0, 1]$, so $B_n = 1$ and condition (ii) reduces to $(\log(mn))^7/n \leq C_2 n^{-c_2}$, i.e. $\log m$ may grow almost as fast as $n^{1/7}$. The band we actually use is *studentised* and is built from the *empirical* (resampling) bootstrap with the standard deviation recomputed in each replicate; validity for exactly this construction — the studentised maximum with either multiplier or empirical bootstrap critical values — is Chernozhukov et al. [15, Thm. 4.3, "Validity of one-step MB and EB methods"], whose Comment 4.6 describes the result as the studentised counterpart of the 2013 theorem. Condition (i) is the one that bites in practice: a hypothesis scoring exactly 1 on every instance has $\hat{\sigma}(c) = 0$. We floor $\hat{\sigma}$ at 10^{-4}; in every experiment reported here we also verify coverage empirically, which is the ultimate check.

(e). The conditional interval is the polyhedral construction of Lee et al. [33]; our implementation restricts the polyhedron to the comparisons the investigator actually made (so that a size-constrained arg-max is handled correctly). The hybrid intersects the conditional interval with a two-sided simultaneous band at level β and then inverts at level $(\alpha - \beta)/(1 - \beta)$; because we use a simultaneous band rather than the exact projection region of Andrews et al. [2], the result is conservative.

(d). Conditionally on the selection half, $\hat{c}$ is fixed and the evaluation half is an i.i.d. sample independent of it, so the $m = 1$ bound applies conditionally and therefore marginally.

B The ratio metric

Write $a_c = \mathbb{E}[\mathrm{LD}(c)]$, $b = \mathbb{E}[\mathrm{LD}(\emptyset)]$, $f = \mathbb{E}[\mathrm{LD}(\text{full})]$ and $F(c) = (a_c - b)/(f - b)$. The map $(a, b, f) \mapsto (a - b)/(f - b)$ is differentiable wherever $f \neq b$, with gradient

$$\left(\tfrac{1}{f-b},\ \tfrac{-1}{f-b} + \tfrac{a-b}{(f-b)^2},\ \tfrac{-(a-b)}{(f-b)^2} \right).$$

Substituting the empirical means and multiplying by the centred per-instance vector gives the influence contribution quoted in Section 4.1:

$$\psi_i(c) = \frac{\big(\mathrm{LD}_i(c) - \hat{a}_c\big) - \big(\mathrm{LD}_i(\emptyset) - \hat{b}\big)}{\hat{f} - \hat{b}} - \hat{F}(c)\, \frac{\big(\mathrm{LD}_i(\text{full}) - \hat{f}\big) - \big(\mathrm{LD}_i(\emptyset) - \hat{b}\big)}{\hat{f} - \hat{b}}.$$

The delta method gives $\hat{F}(c) - F(c) = n^{-1} \sum_i \psi_i(c) + o_P(n^{-1/2})$ uniformly over c under bounded fourth moments. The band we report resamples the instances and recomputes F, its influence function and its standard error in every replicate — the ratio analogue of the bootstrap-t of Section 4(c). Because $\psi_i(c) = \big(u_i(c) - F(c)\, v_i\big)/\bar{v}$ with $u_i(c) = \mathrm{LD}_i(c) - \mathrm{LD}_i(\emptyset)$ and $v_i = \mathrm{LD}_i(\text{full}) - \mathrm{LD}_i(\emptyset)$, each replicate needs only three weighted sums of the $n \times m$ matrices u, u^2 and uv, so the whole band costs three mat-muls per batch of replicates.

A finite-sample alternative. If the per-instance logit differences are known to lie in $[\ell, u]$, one can build two-sided empirical-Bernstein intervals for a_c (simultaneously over c, at level $\alpha/3$), for b and for f (each at level $\alpha/3$), and then report the worst-case value of $(a - b)/(f - b)$ over the resulting box. The result is a finite-sample, simultaneous, post-selection-valid bound; it is considerably more conservative than the bootstrap band, but it does not rely on asymptotics. This is implemented as `psf.functional.ratio_union_lcb`.

C Experimental details

Models. All tiny transformers are pre-LayerNorm decoder-only transformers with learned positional embeddings, trained with AdamW (batch 512, gradient clipping at 1.0, linear warm-up). Every model reaches 100% accuracy on its task; held-out accuracies are:

- TT-A, hierarchical equality, 2 layers × 4 heads, $d = 64$: accuracy 1.0000
- TT-A-IIT, same architecture, alignment planted by IIT: accuracy 1.0000
- TT-B, induction, attention-only, 2 layers × 4 heads, $d = 64$: accuracy 1.0000
- TT-C, three-term modular arithmetic, 3 layers × 4 heads, $d = 64$: accuracy 1.0000
- tt_c_arithmetic_iit: accuracy 1.0000, interchange accuracy at the planted site 1.000
- TT-D, induction, 4 layers × 8 heads + MLPs, $d = 128$: accuracy 1.0000

Hypothesis classes. For the alignment experiments the class is all (site, position, basis, coordinate block) combinations, with sites `resid_pre.0` and `resid_post.`ℓ for each layer ℓ, every token position, two bases (the standard one and one fixed random orthonormal rotation), and contiguous blocks of 8, 16, 32 and 64 coordinates. This class is fixed before any data is seen, which is what the simultaneous bounds require. For GPT-2 the class is the $2^7 = 128$ unions of the seven functional head groups of Wang et al. [49], plus size-stratified random subsets of the 26 published heads and of all 144 heads, plus the published circuit and the full model: 840 distinct circuits in total.

Ablation scheme. Attention heads outside a circuit are mean-ablated: the head's output slice of the input to `c_proj` is replaced by its mean over the ABC distribution (three distinct names), computed *per template and per token position* so that positions stay aligned. MLPs, embeddings and layer norms are never ablated. This is the node-level scheme used by automated circuit discovery [16]; Wang et al. [49] additionally restrict each circuit head to a single token position, which is why our recovered fractions are not numerically identical to theirs.

Positive controls. Two of the five models are trained with interchange intervention training [24] so that a specific subspace is a faithful carrier of a specific high-level variable by construction, giving a known-correct member of the search space. TT-A-IIT plants $V_1 = \mathbf{1}\{a = b\}$ in the first 16 coordinates of the residual stream after layer 0 at the position of token b; TT-C-IIT plants the intermediate sum $S_1 = (a + b) \bmod 10$ in the first 32 coordinates after layer 1 at the final position. Both reach 100% task accuracy and 100% interchange accuracy at the planted site. One implementation note that cost us a day: the source-side forward pass must remain attached to the autograd graph. If the source activations are detached before being patched in, the network receives no gradient asking it to *encode* the variable in the planted subspace — only to react to whatever is already there — and the interchange accuracy never leaves chance.

Compute. The GPT-2 score matrix required 5,058,000 forward passes (204 minutes on one Apple M2 Pro GPU). The greedy search required a further 1,776 circuit evaluations (14 minutes). Tiny-model training and score matrices take under 40 minutes in total. Every statistical result in the paper is then computed from cached score matrices on CPU in a few minutes.

D Additional results

Table 3: Which correction to use, as a function of the search. Simulated binary faithfulness scores with $\theta(c) \equiv 0.80$ for all hypotheses (the complete null), $n = 1000$ interventions, equicorrelation ρ between hypotheses, 400 replications. $\hat{q}$ is the bootstrap critical value and m_{eff} the number of independent hypotheses it corresponds to; both fall far below the Bonferroni value once the hypotheses are correlated, which is why the bootstrap band certifies more than a union bound. Sample splitting overtakes it only when the class is large and uncorrelated.

			naive	multiplicity			certified $L(\hat{c})$			
$\lvert\mathcal{C}\rvert$	ρ	optimism	cover	$\hat{q}$	Bonf.	m_{eff}	boot	union	split	hybrid
20	0.0	0.023	0.38	2.95	2.81	17.0	0.788	0.765	0.771	0.778
20	0.5	0.020	0.54	2.89	2.81	10.9	0.784	0.762	0.771	0.775
20	0.9	0.013	0.76	2.62	2.81	8.1	0.780	0.755	0.770	0.769
200	0.0	0.034	0.00	3.72	3.48	926.0	0.790	0.767	0.771	0.783
200	0.5	0.029	0.13	3.60	3.48	154.1	0.786	0.762	0.769	0.777
200	0.9	0.020	0.50	3.13	3.48	28.6	0.782	0.753	0.770	0.771
2000	0.0	0.042	0.00	4.38	4.06	3231.6	0.792	0.767	0.770	0.785
2000	0.5	0.036	0.02	4.22	4.06	2117.8	0.786	0.761	0.771	0.778
2000	0.9	0.024	0.37	3.55	4.06	94.4	0.781	0.749	0.769	0.770

Table 4: When the target population is new *templates*. Prompts are drawn template-by-template — the templates themselves are resampled — which is the honest model of how an IOI evaluation set is produced. The selection is the best circuit of at most eight heads (the unconstrained arg-max is perfect on every sampled prompt, and its degeneracy would swamp the comparison). The finite-sample union bounds assume i.i.d. instances and are therefore omitted: they are not valid under this sampling scheme. Median intra-template correlation of per-instance scores: 0.055 over $G = 30$ templates, giving a design effect of 4.6 at $n = 2000$.

	coverage			certified $L(\hat{c})$		
method	$n = 500$	$n = 1000$	$n = 2000$	$n = 500$	$n = 1000$	$n = 2000$
naive $\pm z\,\text{SE}$	0.814	0.760	0.726	0.765	0.771	0.777
bootstrap simultaneous band	0.997	0.997	0.954	0.695	0.727	0.750
cluster bootstrap band	1.000	0.997	1.000	0.678	0.695	0.706

Table 5: An adaptive search on GPT-2 small. A greedy pruner removes attention heads one at a time using $n = 200$ interventions, issuing 1,776 queries in total, and stops at 11 heads. On its own search set the returned circuit scores 0.917; on 1200 held-out interventions its true score is 0.916. Only the last three rows are valid after an adaptive search.

what is reported	value	status
naive bound on the search data	0.860	**invalid**: the circuit was chosen on these interventions
uniform over the 10^{18} circuits the search could have reached	0.601	valid, but vacuous from the search data alone
held-out, 200 fresh interventions	0.866	valid
held-out, 1200 fresh interventions	0.894	valid

Table 6: Why the bootstrap must recompute the standard deviation. Binary faithfulness scores make the sample mean and the sample standard deviation dependent, so a multiplier bootstrap that holds $\hat{\sigma}$ fixed understates the upper tail of the studentised maximum and under-covers. Resampling the interventions and recomputing $\hat{\sigma}$ in every replicate restores the nominal level at a small cost in width. All entries over 500 replications; nominal level 0.95.

				coverage		width	
$\lvert\mathcal{C}\rvert$	n	ρ	θ	fixed $\hat{\sigma}$	bootstrap-t	fixed $\hat{\sigma}$	bootstrap-t
60	400	0.0	0.75	0.886	0.950	0.063	0.068
60	400	0.6	0.75	0.892	0.944	0.061	0.066
300	400	0.0	0.75	0.864	0.966	0.070	0.077
60	1500	0.0	0.75	0.910	0.942	0.034	0.035
60	400	0.0	0.95	0.576	0.966	0.025	0.037
60	400	0.0	0.5	0.954	0.966	0.078	0.079